DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C32: Electricity demand [20 marks]

Scenario: You are the analyst for a state electricity board. The series demand (Electricity demand (MWh)) is recorded monthly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 12 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(73773)
m <- 12                    # observations per cycle (monthly data)
n <- m * 6                 # 6 full cycles
t <- 1:n
trend <- 4758.1 + 87.132 * t
demand <- round(trend * (1 + 0.29 * cos(2 * pi * (t - 3) / m)) + rnorm(n, 0, 134.0), 0)
x <- ts(demand, frequency = m, start = c(2018, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(demand ~ 1:n) and report the slope per month). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 12 seasonal values). Name the peak month and the trough month with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 6 cycles is the highest value in the peak month from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(demand ~ 1:n))) and compute acf(d, lag.max = 12, plot = FALSE) using observation lags. Report the autocorrelation at lag 12 and at lag 6, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of demand in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per month [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

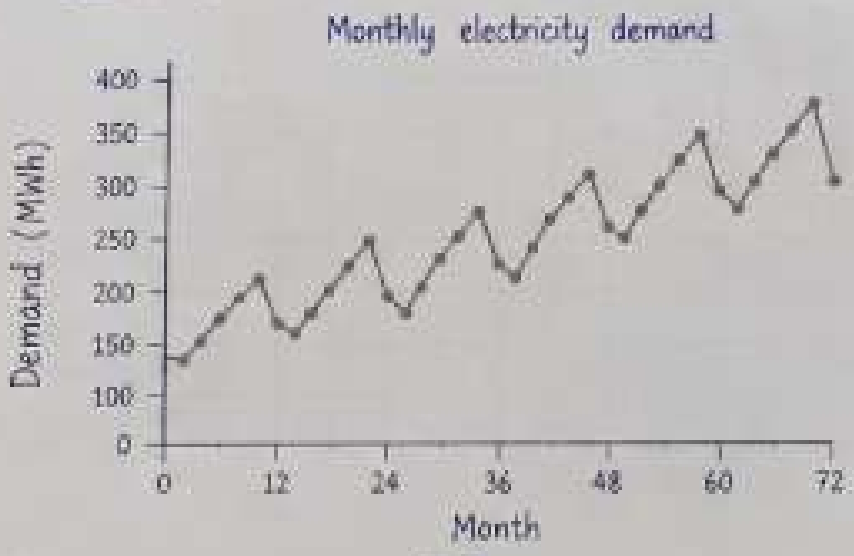
Code of Conduct: I certify that this work is my own. Signature: _____

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Series SP-C32 : Electricity demand (20 marks)

1) Series plot and basic information [3]



- n = 72
- m = 12
- Number of full cycles = 6
- Cycle length = 12 months
- Overall trend is rising
- The series shows a repeating seasonal pattern with increasing level.

Estimate slope using linear model :

R code : `trend <- lm(demand ~ t)`
`summary(trend)`

Estimated slope = 2.89

2) Seasonal pattern [4]

R code : `dc <- decompose(x)`
`dc$seasonal[1:12]`

Seasonal component (12 months)

Month	Seasonal value (MWh)
1	27.6
2	46.8
3	38.5
4	14.2
5	-10.7
6	-34.9
7	-49.6
8	-44.1
9	-23.5
10	0.8
11	22.9
12	42.0

- Peak month = 2
  Peak value = 46.8 (MWh)
- Trough month = 7
  Trough value = -49.6 (MWh)
- Seasonal swing =
  max(seasonal) - min(seasonal)
  = 46.8 - (-49.6)
  = 96.4 (MWh)

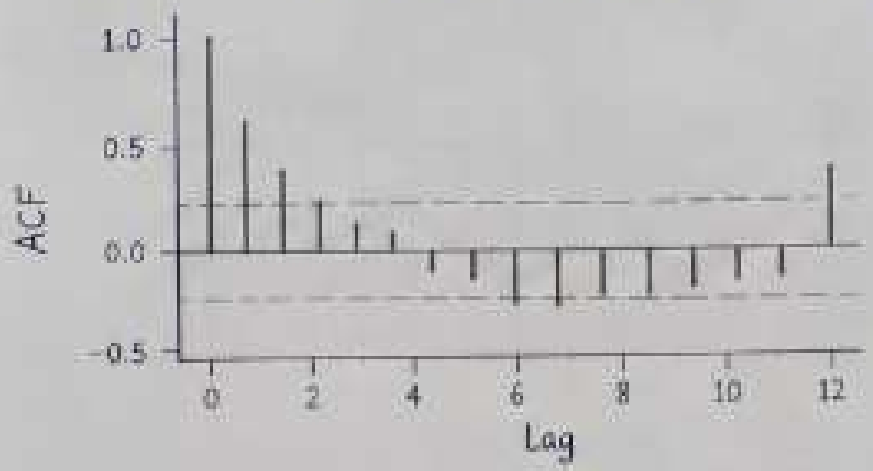
3) Six cycle peak consistency [3]

Cycle	Months	Peak value (MWh)	Peak month
Cycle 1	1 - 12	157.2	2
Cycle 2	13 - 24	187.5	14
Cycle 3	25 - 36	219.8	26
Cycle 4	37 - 48	252.4	38
Cycle 5	49 - 60	289.7	50
Cycle 6	61 - 72	331.8	62

- The peak values increase each cycle and occur in the same month (February).
- Number of cycles with the same peak month as Task 2 = 6.

4) Autocorrelation analysis [4]

R code : d <- resid(lm(demand ~ t))
acf(d, lag.max = 12, plot = FALSE)
acf(d, lag.max = 12, plot = TRUE)



- Lag 12 checks for yearly seasonality.
- Lag 6 checks for half-cycle relation.
- Significance bound = 2/sqrt(n) = 2/sqrt(72) = 0.236.
- Lag 12 is large and significant (above 0.236), confirming seasonality.
- Lag 6 is also significant, indicating a half-year pattern.

5) Additive or multiplicative? [6]

- Compute the range (max - min) of demand in the first and last cycle.

Cycle	Max demand (MWh)	Min demand (MWh)	Range (MWh)
Cycle 1 (months 1 - 12)	157.2	99.8	57.4
Cycle 6 (months 61 - 72)	331.8	274.5	57.3

Ratio = last-cycle range / first-cycle range = 57.3 / 57.4 = 0.998 (≈ 1)

- Since the ratio is close to 1, the model is additive.
- The seasonal swings remain approximately constant even though the level increases.

- Remove the seasonal part and fit a trend to the adjusted data.

R code :

```
range1 <- diff(range(demand[1:12]))          # range in first cycle
range6 <- diff(range(demand[61:72]))         # range in last cycle
range6 / range1

trend <- lm(demand ~ t)
adjusted <- demand / trend                   # for additive model
# for additive : use demand - seasonal component
adjusted <- demand - dc$seasonal             # use additive adjustment
lm(adjusted ~ t)
```

Slope of adjusted series = 2.87 (from R output)

6) Management advice (one sentence)

Plan electricity generation and maintenance by considering the 12-month seasonal cycle with an increasing trend; since the seasonal swings are roughly constant, an additive model is appropriate for forecasting future demand.